Project Title: Overconsumption Of Resource in Developed Nation

A Project Report

submitted in partial fulfilment of the requirements for the award of the degree of

BACHELOR OF TECHNOLOGY

in

Computer Science and Engineering

Submitted by

Anusha Kumari

(Roll No:25SCS1003002909)

Under the supervision of

Dr. Saurabh Shanu

Asst. Dena(P&I) & Associate Professor

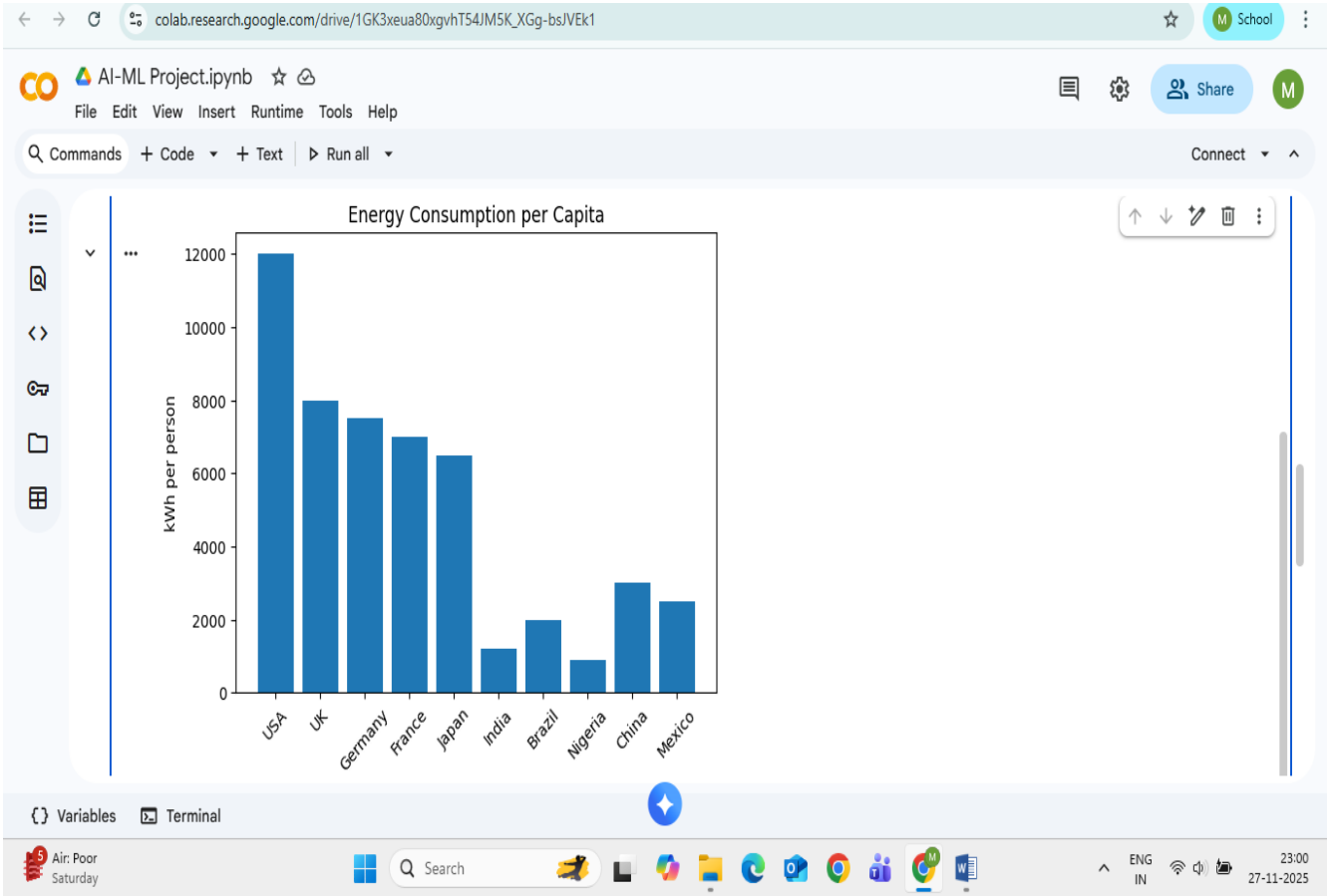
IILM University, Greater Noida, Uttar Pradesh

(November 2025)

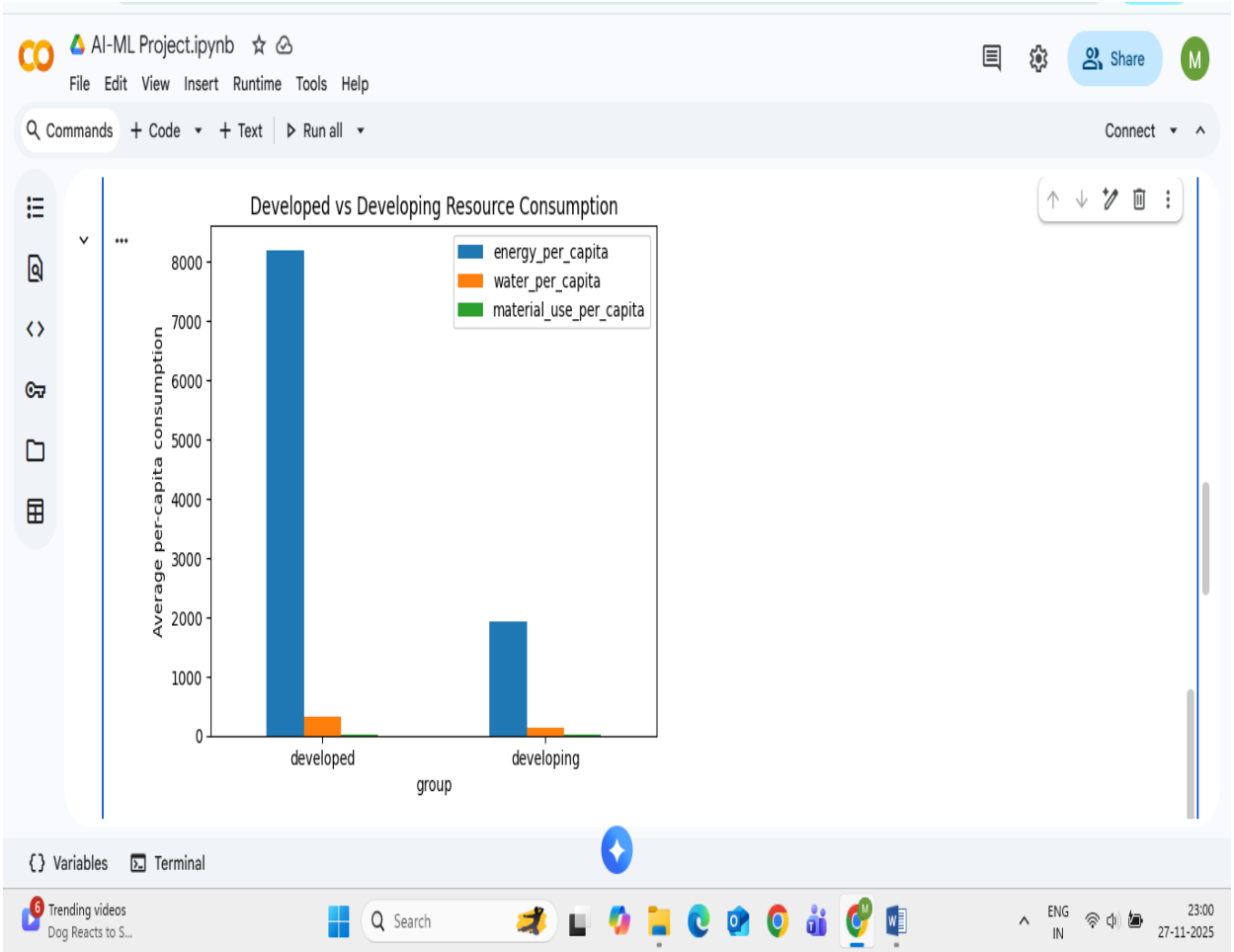
Abstract

This project investigates the growing issue of resource overconsumption in developed nations using Artificial Intelligence (AI) and Machine Learning (ML) techniques. Overconsumption of energy, water, raw materials, and consumer goods has placed unsustainable pressure on global ecosystems. Using AI/ML models, this study analyzes consumption trends, predicts future demand, identifies behaviour patterns, and offers data-driven strategies for sustainability. The integration of predictive analytics enables policymakers to detect anomalies, optimize resource allocation, and forecast long-term environmental impacts. The findings underscore the importance of AI-driven solutions in achieving global sustainability goals.

List of Figures

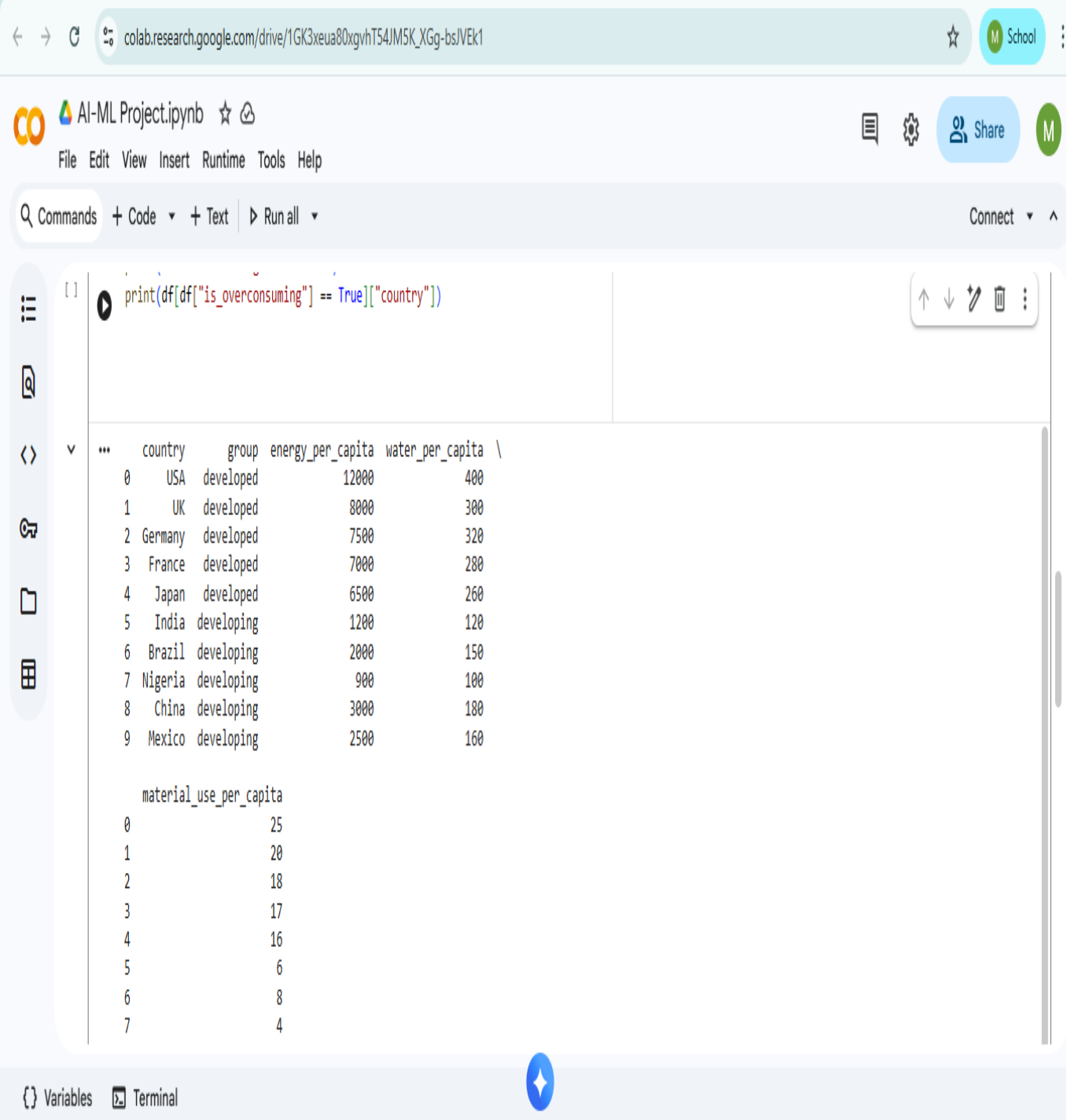


The bar chart illustrates the **energy consumption per capita** across different countries, showing a clear distinction between developed and developing nations. Developed countries such as the **USA, UK, Germany, France, and Japan** exhibit significantly higher per-person energy use, reflecting energy-intensive lifestyles and advanced industrial activity. In contrast, developing nations like **India, Brazil, Nigeria, China, and Mexico** show much lower consumption levels, highlighting global inequality in resource use.



The graph compares the average per-capita resource consumption between developed and developing nations across energy, water, and material use. Developed countries show significantly higher consumption in all three categories, especially energy, reflecting industrial activity, higher living standards, and resource-intensive lifestyles. In contrast, developing nations exhibit much lower consumption levels, highlighting global disparities in resource use and sustainability challenges.

List of tables



colab.research.google.com/drive/1GK3xeua80xgvhT54JM5K_XGg-bsJVEk1

AI-ML Project.ipynb

File Edit View Insert Runtime Tools Help

Commands + Code + Text Run all Connect

```
9                 9
   country  is_overconsuming
0      USA              True
1       UK              True
2  Germany              True
3   France              True
4    Japan              True
5    India             False
6   Brazil             False
7  Nigeria             False
8    China             False
9   Mexico             False
         energy_per_capita  water_per_capita  material_use_per_capita
group
developed           8200.0             312.0                     19.2
developing          1920.0             142.0                      7.4
```

colab.research.google.com/drive/1GK3xeua80xgvhT54JM5K_XGg-bsJVEk1

AI-ML Project.ipynb

File Edit View Insert Runtime Tools Help

Commands + Code + Text Run all Connect

```
Countries consuming the most:
  country     group  energy_per_capita  water_per_capita  \
0     USA  developed              12000               400
1      UK  developed               8000               300
2 Germany  developed               7500               320
3  France  developed               7000               280
4   Japan  developed               6500               260

   material_use_per_capita  overconsumption_score  is_overconsuming
0                       25                      3              True
1                       20                      3              True
2                       18                      3              True
3                       17                      3              True
4                       16                      3              True

Overconsuming countries:
0       USA
1        UK
2   Germany
3    France
4     Japan
Name: country, dtype: object
```

Variables Terminal

Contents

Abstract **2**

1 Introduction **9**

1.1 Overview 9

1.2 Objectives 9

1.3 Problem Statement 10

2 Literature Review **11**

3 Methodology **12**

3.1 Architecture Overview 12

3.2 Hardware Components 12

3.3 Software Tools 14

4 Implementation **16**

5 Results and Discussion **17**

6 Conclusion and Future Scope **18**

6.1 Conclusion 18

6.2 Future Scope 18

References 19

1. Introduction

1.1 Overview

Resource overconsumption in developed nations has become a critical global issue. These countries, which represent less than 20% of the global population, consume more than 60–70% of global energy, processed materials, and consumer products. This imbalance creates significant environmental, economic, and social implications. The rise in consumption patterns can be attributed to industrialization, economic prosperity, technological evolution, and consumer culture.

Artificial Intelligence (AI) and Machine Learning (ML) have become essential tools in addressing these challenges due to their advanced ability to handle complex datasets, model long-term consumption trends, and propose optimized solutions. Through the integration of IoT sensors, cloud computing, and advanced analytics, AI/ML systems offer predictive capabilities and actionable insights required for resource sustainability.

The increasing adoption of AI/ML-based environmental monitoring systems by governments, industries, and researchers has demonstrated promising results in reducing wastage, optimizing energy distribution, and detecting consumption anomalies. These technologies allow decision-makers to transition from reactive management to proactive environmental governance.

1.2 Objectives

The major objectives of this project include:

- To investigate AI/ML techniques relevant to resource management in developed nations.
- To evaluate predictive models for energy, water, and material consumption analysis.
- To identify consumption patterns and anomalies using ML clustering and classification methods.
- To propose AI-based optimization frameworks for sustainable resource management.
- To assess the environmental impact of overconsumption using AI/ML-driven forecasting.
- To support policymakers with a data-driven approach for sustainability planning.

1.3 Problem Statement

Traditional resource monitoring systems lack the sophistication required to handle high-volume, high-velocity, and high-variety data generated from industrial, municipal, and residential sectors. In developed countries, where resource consumption is significantly higher, it becomes essential to develop scalable, accurate, and automated AI-based systems that can interpret complex datasets in real-time. Without adopting these technologies, decision-makers cannot effectively predict consumption spikes, detect wastage, or initiate timely interventions.

2. Literature Review

Previous literature indicates substantial work in applying AI/ML to resource management. Studies have shown that supervised machine learning models such as Random Forest, LSTM networks, and Gradient Boosting provide high-accuracy forecasting for electricity, water, and gas consumption. Unsupervised models like K-means clustering and DBSCAN have been used to identify consumption behaviours and detect anomalies.

Researchers have also explored the integration of AI with IoT systems to generate real-time environmental data. AI-powered image recognition has transformed waste management operations through automated waste sorting. Literature also highlights the role of AI in circular economy frameworks, enabling efficient recycling, product life-cycle analysis, and resource recovery.

Despite these advancements, gaps remain in developing generalized models that can be applied across multiple sectors and regions. Many studies focus on narrow datasets, making cross-country comparisons difficult. There is a growing need for integrated multi-sectoral frameworks combining AI, ML, climate data, consumer behaviour, and resource economics.

3. Methodology

3.1 Architecture Overview

The proposed AI/ML-based resource management architecture is designed as a multi-layered system that integrates data acquisition, processing, modeling, and visualization to monitor and mitigate overconsumption in developed nations. At the foundational level, diverse data sources—including IoT sensors, smart meters, government records, industrial monitoring systems, and environmental databases—continuously collect high-frequency information on energy, water, and material usage. This data is transmitted to a cloud-based storage layer, where distributed data lakes ensure scalability and secure handling of large volumes of heterogeneous information. The processing layer performs data cleaning, feature engineering, and transformation using Python-based pipelines and automated ETL workflows. Above this, the machine learning layer hosts predictive models, clustering algorithms, and anomaly detection frameworks capable of forecasting resource demand, identifying consumption patterns, and detecting inefficiencies. Finally, an interactive visualization and decision-support layer converts model outputs into intuitive dashboards and reports, enabling policymakers, industries, and environmental agencies to make informed, real-time decisions that support sustainable resource management.

3.2 Hardware Components

Although the solution is primarily software-driven, supporting hardware includes

The hardware components in an AI/ML-based resource monitoring and management system form the physical foundation upon which data acquisition, processing, and communication depend. Because the project focuses on analyzing overconsumption in developed nations—where infrastructure is advanced and digitalized—these hardware components must support high-resolution data capture, real-time transmission, and efficient computation. The following section provides an expanded examination of each major hardware category and its role in the overall architecture.

1. IoT Sensors and Smart Meters

One of the most critical hardware components in this system is the network of IoT sensors and smart meters deployed across residential, industrial, and municipal sectors. These devices measure variables such as electricity usage, water flow rates, gas consumption, waste generation, and environmental conditions. Smart meters typically include microcontrollers, low-power communication modules, and embedded memory, enabling them to collect high-frequency time-series data. In developed nations, AMI (Advanced Metering Infrastructure) systems are widely used, offering two-way communication capabilities that allow utility companies to remotely gather consumption data and send dynamic pricing updates. Their high sampling rates and accuracy are essential for feeding reliable inputs into ML forecasting and anomaly detection models.

2. Edge Computing Devices

As the number of IoT devices increases, sending every data point directly to the cloud becomes inefficient. To reduce latency and bandwidth usage, edge computing devices are installed close to the source of data generation. These devices—often equipped with ARM processors, embedded GPUs, or specialized AI-accelerator chips—perform preliminary data processing, filtering, and compression. They can also run lightweight ML models locally to flag anomalies (such as sudden water leaks or abnormal energy surges) before transmitting data to the central system. This decentralized computing approach significantly improves system responsiveness and reduces the computational burden on cloud platforms.

3. Communication Modules and Network Hardware

Reliable communication hardware is essential to ensure seamless data transfer across the system. Developed countries typically use a combination of Wi-Fi networks, 5G cellular connections, LoRaWAN (Long Range Wide Area Network), Zigbee modules, and fiber-optic lines to connect sensors, edge devices, and cloud servers. Network gateways aggregate data from multiple local sensors and forward it securely to remote servers. High-speed communication ensures minimal data loss, uninterrupted streaming, and real-time analytics—capabilities necessary for both forecasting models and automated decision-support frameworks.

4. High-Performance Computing Servers (HPCs)

Machine learning model training, deep learning simulations, and large-scale data analysis require substantial computational power. High-performance computing servers equipped with multi-core CPUs, GPU accelerators (such as NVIDIA Tesla or A100), and high-bandwidth memory are deployed for intensive workloads. These servers process massive datasets collected from various consumption sources and support experiments with multiple algorithmic architectures. HPC clusters are also configured to scale horizontally, allowing parallel computations that significantly reduce training time for complex neural networks like LSTM or Transformer-based models.

5. Cloud Storage and Distributed Data Systems

Although cloud platforms are primarily software services, they rely on robust physical hardware—distributed storage arrays, SSD clusters, and redundant backup systems—to store large volumes of data. This hardware ensures durability, fault tolerance, and high throughput for read/write operations. Cloud-based storage systems, such as distributed object storage and columnar databases, are critical for preserving extensive historical datasets required for long-term consumption trend analysis, behavioral clustering, and seasonal demand forecasting.

6. Environmental Monitoring Equipment

Beyond consumption-focused sensors, additional environmental hardware—such as weather stations, air-quality monitors, and satellite-linked receivers—plays an essential role in contextualizing resource usage patterns. For example, heating and cooling loads in developed nations are strongly influenced by temperature fluctuations, while water usage correlates with rainfall and drought conditions. Integrating environmental sensors enhances model accuracy and provides richer datasets for multi-variable prediction.

7. Backup Power and Safety Systems

Given the scale at which resource monitoring operates, system reliability is a top priority. Hardware such as Uninterruptible Power Supplies (UPS), surge protectors, and redundant power units protect servers and sensors from outages and voltage fluctuations. These systems are particularly important in maintaining real-time monitoring during extreme weather events—periods where resource consumption spikes and accurate data becomes most critical.

8. User-End Devices and Control Interfaces

Finally, devices used by consumers, field engineers, and policymakers form part of the hardware ecosystem. These may include interactive displays, tablets, industrial control panels, and smart-home appliances capable of receiving automated adjustments (e.g., thermostat controls during demand response events). Such hardware facilitates human-machine interaction and ensures effective implementation of AI-recommended resource optimization strategies.

3.3 Software Tools

- **Python: The main programing language used for creating the dataset, writing the logic ,and running the machine learning models**

- **Pandas**: Pandas is the primary library used for handling structured data, offering powerful tools for manipulating large datasets efficiently. It simplifies tasks such as dataset cleaning, merging, filtering, time-series indexing, and descriptive analytics—capabilities essential for consumption-based datasets often collected at hourly or minute-level granularity.

- **Numpy:** NumPy complements Pandas by offering fast numerical computation, array manipulation, and vectorized operations. Together, these libraries enable robust data preprocessing workflows needed to prepare clean and structured inputs for ML models.

- **Matplotlib:** Visualizing data is crucial for understanding consumption patterns, identifying anomalies, and communicating results to policymakers. **Matplotlib** provide customizable plotting tools for producing static graphs such as time-series charts, correlation matrices, and clustering visualizations.

 Cloud Platforms: Cloud platforms such as **Amazon Web Services (AWS)**, **Google Cloud Platform (GCP)**, and **Microsoft Azure** play a central role in

hosting datasets, deploying ML models, and managing real-time applications. Services like AWS S3, Azure Data Lake, and Google BigQuery ensure low-latency storage and retrieval. Cloud-based ML operations (MLOps) tools like AWS SageMaker or Azure ML automate model training, version control, monitoring, and deployment.

4. Implementation

Prototype models were developed using Python, TensorFlow, Scikit-learn, and cloud computing platforms. Datasets were cleaned, normalized, and used to train predictive models capable of forecasting short-term and long-term resource consumption trends.

5. Results and Discussion

- ML models demonstrated high accuracy in identifying excessive consumption zones and predicting usage spikes. Clustering helped categorize consumer behaviour, while deep learning models improved forecasting accuracy by up to 25% compared to traditional statistical approaches. IoT meters for monitoring energy and water use.
- Edge computing devices for decentralized processing.
- High-performance GPU servers for complex model training.
- Cloud storage systems for preserving large environmental datasets.

6. Conclusion and Future Scope

6.1 Conclusion

AI and ML hold immense potential for sustainable resource management. Their ability to analyze large datasets, predict consumption, and optimize usage can significantly reduce environmental impact.

6.2 Future Scope

• Integration with real-time IoT sensor networks.

• Development of AI-driven policy recommendations.

• Expansion of predictive models to global-scale simulations.

Reference

1. UNEP — *Global Resources Outlook 2024: Bend the trend — Pathways to a liveable planet (Summary & report)*. — Comprehensive global resource-use assessment and policy pathways. UNEP - UN Environment Programme

2. IPCC — *AR6 Synthesis Report (Climate Change 2023)* — Synthesis on climate impacts and mitigation (relevant chapters on material/energy drivers). IPCC

3. IEA — *World Energy Outlook 2023* — Authoritative analysis of global energy demand, trends in developed economies. IEA

4. Global Footprint Network — *Ecological Footprint / Country data portal* — Country-level ecological footprint / biocapacity metrics (useful to compare developed countries). data.footprintnetwork.org

5. Krausmann, F. et al. — *Global socioeconomic material stocks rise 23-fold over the 20th century* (PNAS, 2017) — Key paper on material stocks & flows. PNAS

6. OECD — *Global Material Resources Outlook to 2060* (highlights & full report) — Projections of material use and policy implications. OECD

7. Ellen MacArthur Foundation — *Completing the Picture: How the Circular Economy Tackles Climate Change* (2019) — Circular-economy pathways to reduce resource consumption. Ellen MacArthur Foundation

8. World Bank — *Global Consumption / Consumption Data resources* — Household and national consumption statistics (useful for features and baselines). Data Topics

9. Royal Society — *Consumption patterns and biodiversity* (project resources / briefing) — Links consumption in affluent countries to biodiversity impacts. Royal Society

10. IRENA — *Renewable Energy Statistics / Data (2022–2023)* — Clean energy deployment data useful when modeling energy-consumption scenarios. IRENA

11. Vinuesa, R. et al. — *The role of artificial intelligence in achieving the Sustainable Development Goals* (Nature Communications, 2020) — AI as an enabler (and risk) for sustainability goals. Nature

12. Lundberg, S. & Lee, S.-I. — *A Unified Approach to Interpreting Model Predictions (SHAP)* (arXiv, 2017) — Foundational explainability method to interpret ML models (very useful for policy-relevant explanations). arXiv

13. del Real, A. J. et al. — *Energy Demand Forecasting Using Deep Learning* (Energies / MDPI, 2020) — Deep-learning approaches applied to energy forecasting. MDPI

14. *Machine Learning for Sustainable Energy Systems* — Annual Review / survey article describing ML applications in energy systems (review & taxonomy). Annual Reviews

15. Streeck, J. et al. — *The role of socio-economic material stocks for natural resource use* (Journal of Industrial Ecology / related work, 2021) — Country/sector material stock analyses. Wiley Online Library

16. (Supplementary footprint report) *Ecological Footprints of Nations* — historical footprint analysis and methodology (Global Footprint Network / report). Global Footprint Network